

脑智能团队周报

2026年5月24日

周报概述

本周重点工作汇总

本周实际收到 10 份；胡浩琪、廖元请假写本子；张志齐正在回复答辩意见并修改论文；孙嘉璐实验尚未跑完，下次跑完一起提交；张台忍在期末考试和毕设答辩。标注 ★ 表示本周有具体量化实验结果。

- 陈旋达：推进单神经元听故事、MEG 语言/韵律分析与社科基金/digit brain proposal；下周整理 SEEG 范式并继续联合编码/解码。
- 李思睿★：完成 LightSSIR/MADSNet 迭代与消融，SREDS 达 PSNR 37.232、SSIM 0.956、LPIPS 0.028。
- 林哲源★：推进 S3N 与 SpikeTSE，S3N-Condition 2 在 GSC V2 达 96.93%，SpikeTSE 在 AVED 上 SI-SDR=17.31。
- 万俊杰★：推进 BDCTR 微动作识别与 TIP 初稿，MA-52 Codabench 九流融合 test F1_mean=68.07%。
- 张少琛★：完成 DSAINet SNN 化初步消融，BCIC-IV-2a Subject 1 当前最优 All Best ACC=67.19%。
- 张嘉程：整理 UniEEG/EEG MLLM 训练、评估与可视化，Method 主要小节已形成初稿。
- 郑又村：完成北京端一例 sEEG 采集并熟悉流程，定位音频失真问题，继续准备长期采集。
- 邱雯★：完成 EMG 采集与 emg2text 局部证据方案验证，oracle boundary 下 decoded WER=0.2549。
- 汤宓★：完成华山多模态分析报告与 bistable 数据 spike sorting，ATL 初步约 148 个神经元。
- 贾煜晗：完成 M3CV 数据预处理、本科毕设答辩和 TIP rebuttal，下一步测试模型并设计 ACIL 创新点。

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高思翰 未提交

xuanda0523周报

科研周报 | 姓名：陈旋达 | 日期：2025-05-23 | 研究方向：cogsci

1.本周学习/研究内容

- 其实是本月的内容
- 单神经元数据听故事的分析方法比较了三种不同的算法；但是由于数据量限制无法确定哪一种可能达到显著；有一种出现和其他不同的趋势；需要和最新的数据比较后再确定
- 完成社科基金；剩余部分进行修改 25号前提交到系里
- digit brain proposal提交
- MEG的数据分析倒是有不少进展 不仅成功复现了 Gwilliams et al 2025 拓展到了prosodic decoding；接下来会考虑联合联合编码的docoding以及基于互信息的表征decoding；同时下载了法语、德语以及荷兰语（还未下载成功）的听故事数据集准备做跨语言的测试泛化

属性	MEG-MASC	MEG-SCANS	ds007523	DSC_3011085	DSC_3027007
语言	English	German	French	English	Dutch
被试数	24 (有效)	24 (sub-01 pilot)	58	3 × 10 sessions	30
总时长/人	~2h	~1h	~1.5h	~10h	~1h
故事数	4 篇短篇	2 本有声书	1 本 (9段)	1 本有声书	1 本有声书
MEG 设备	KIT	Elekta Neuromag	Elekta Neuromag	CTF	CTF
通道数	208 axial grad	306 (102mag+204grad)	306 (102mag+204grad)	275	~275
采样率	1000 Hz	1000 Hz	1000 Hz	2400 Hz	1000 Hz
MRI	少数	全部有 T1	全部有 T1	未知	未知
标注	events.tsv (词+音素)	Praat TextGrid (词+音素)	events.tsv (词级别)	events.tsv	待确认
数据量	~200 GB	~100 GB	448 GB	242 GB	265 GB (下载中)
论文	Gwilliams 2025	Habersetzer 2025	d'Ascoli 2025	Armeni 2022	Martin
状态	✅ 完整分析	✅ 完整分析	✅ 下载完成	⚠️ 缺 stimuli	⌚ 下载中

3.下周学习/研究计划

- 提交社科基金
- SEEG 范式整理 准备收集新的宏电极数据；prosody- syntax interface项目放到SEEG；声调 encoding项目提上来；听故事数据有必要；pronoun resolution 解码也许可以？
- 看看MEG的linguistic+prosodic联合编码的docoding 能不能完成

4.其他事项/需求 (可选)

胡浩琪 未提交

Attention-Guided Convolutional Spiking Neural Networks for Spike Camera Image Reconstruction

Abstract

Spike cameras are bio-inspired sensors that output asynchronous binary spike streams at ultra-high temporal resolution, posing unique challenges for image reconstruction. Existing SNN-based approaches, while energy efficient, suffer from limited representational capacity stemming from information loss inherent in discrete spike activity. In this paper, we propose **EnhancedSSIR**, a novel SNN-based spike camera image reconstruction network that, for the first time, systematically adapts attention mechanisms and large-kernel convolutional attention from the image super-resolution domain into the neuromorphic reconstruction setting. Specifically, we introduce a **Spike Attention Block (SAB)** that fuses dual-path spiking features through a Hybrid Dimension Attention module (HDA) for joint temporal-channel recalibration, and a Large-Kernel Convolutional Attention module (LKA) that emulates self-attention's long-range modeling with a shared large kernel and instance-dependent dynamic kernels—avoiding costly softmax attention entirely. We further adopt an **IbraLIF** neuron model, a multi-level quantized spiking neuron that enriches the representational range of binary spike signals beyond the conventional 0/1 regime. An encoder-decoder architecture with gradient checkpointing enables scalable training under memory constraints. Experiments on the SREDS benchmark demonstrate that EnhancedSSIR achieves competitive reconstruction quality compared to state-of-the-art methods while maintaining the energy efficiency characteristic of SNN computation.

1. Introduction

Spike cameras represent a paradigm shift in bio-inspired sensing: unlike conventional frame-based cameras or event cameras, a spike camera's pixels independently accumulate photons and fire a binary spike upon crossing a threshold, producing a continuous stream of spatiotemporal binary data. Reconstructing intensity images from such streams is a fundamental yet challenging problem that demands models capable of processing high-frequency temporal information efficiently.

Deep learning approaches have dramatically improved reconstruction quality. The state-of-the-art method Spk2ImgNet [Zhao et al. 2021] uses deformable convolution-based feature alignment and achieves impressive results, yet demands substantial memory (8.77 GB) and latency (143.2 ms per frame). SSIR [Zhao et al. 2024] demonstrates that Spike-Element-Wise (SEW) residual SNNs can reach comparable performance at roughly 1/1000 the energy cost, opening the door to neuromorphic deployment. However, the SSIR architecture is shallow and relies on basic bottleneck convolutions, leaving substantial headroom in representational power.

Meanwhile, the image super-resolution (SR) community—a closely related pixel regression task—has witnessed remarkable advances through attention mechanisms and large-kernel convolutions. SpikeSR [Xiao et al. 2025] shows that SNN-based attention blocks can exploit the active learning state of spiking neurons to surpass ANN-based SR methods in remote sensing. ESC [Lee et al. 2025] demonstrates that a shared large kernel combined with dynamic kernels can efficiently emulate the instance-dependent weighting of self-attention, eliminating memory-bound score matrix operations.

We argue that these advances are directly transferable to spike camera reconstruction. Both tasks involve pixel-level regression from degraded, high-frequency signals, and both benefit from long-range dependency modeling and efficient feature recalibration. However, existing neuromorphic reconstruction models have not yet incorporated these insights.

In this paper, we present **EnhancedSSIR**, which makes the following contributions:

- **We adapt the Hybrid Dimension Attention (HDA)** paradigm from SNN-based SR to spike camera reconstruction, realizing joint temporal-channel recalibration that modulates membrane potentials for superior spike feature representation in the reconstruction setting.
- **We introduce Large-Kernel Convolutional Attention (LKA)** into SNN-based reconstruction, combining a network-wide shared large kernel with per-sample dynamic kernels to capture long-range dependencies with memory efficiency far superior to windowed self-attention.
- **We use the IbraLIF neuron**, a multi-level quantized spiking neuron that moves beyond binary spikes to a fine-grained discrete coding space, enriching the information capacity of each layer while retaining SNN's event-driven computation properties.
- **We design an encoder-decoder macro-architecture** with gradient checkpointing, enabling the first hierarchical SNN reconstruction network scalable to high-resolution spike streams.

Extensive experiments on the SREDS benchmark validate that EnhancedSSIR substantially improves upon the SSIR baseline and achieves favorable accuracy-efficiency trade-offs against both ANN and SNN reconstruction methods.

2. Related Work

Spike Camera Image Reconstruction

Spike camera reconstruction has been studied via temporal filtering (TFP, TFI [Zhu et al. 2019]), neuronal models (TFSTP [Zheng et al. 2021]), and deep learning (Spk2ImgNet [Zhao et al. 2021]). SSIR [Zhao et al. 2024] first demonstrated SEW-based SNNs for spike reconstruction, achieving significant energy savings through a bottleneck architecture and a Progressive Filtering Module (PFM) for binary-to-continuous conversion. Our work builds directly on SSIR, extending its representational capacity with attention and large-kernel mechanisms.

Spiking Neural Networks for Vision

SNNs have been applied to image classification [Yao et al. 2023], object detection [Kim et al. 2020], and optical flow estimation [Paredes-Vallés et al. 2019]. For image restoration, SpikeSR [Xiao et al. 2025] introduced Spiking Attention Blocks for remote sensing super-resolution, showing that hybrid CNN-SNN attention effectively mitigates spiking information loss. Our work applies analogous principles to the spike camera reconstruction task, which differs fundamentally in its binary input modality and the need to recover absolute luminance rather than upscale existing textures.

Attention and Large-Kernel Convolution for Image Restoration

Transformer-based SR methods [SwinIR, SRFormer, ATD] leverage self-attention's long-range modeling, but their memory overhead (from materializing the score matrix) is prohibitive for large feature maps. ESC [Lee et al. 2025] addresses this with a Convolutional Attention module that uses a shared 13×13 large kernel and layer-specific dynamic depth-wise kernels to emulate self-attention without materializing any score matrix. We adapt this design principle into the SNN domain, replacing conventional neurons with IbraLIF and integrating it within our SEW residual framework.

Multi-Level Spike Encoding

Standard LIF neurons produce binary outputs, limiting representational capacity. IBRA-based quantized neurons [SIBRA formulation] extend discrete firing to multi-level representations by scaling and rounding activations within a bounded range, providing a continuous-approximation-friendly encoding that improves learning stability while retaining interpretable spike semantics.

3. Proposed Method

3.1 Overview

The overall architecture of EnhancedSSIR is illustrated in Figure 1. Given an input spike sub-stream of 41 binary frames $\mathbf{x} \in \{0,1\}^{(41 \times H \times W)}$, the network maps it to a reconstructed

grayscale image $\hat{\mathbf{I}} \in [0,1]^{(H \times W)}$. The pipeline consists of four stages:

1. **Stem**: A shallow feature extractor using a spiking convolution followed by a lightweight Spike Attention Block.
2. **Encoder**: A stride-2 spiking convolution that halves spatial resolution, followed by two deep Spike Attention Blocks with Large-Kernel Attention.
3. **Bottleneck**: A single Spike Attention Block with LKA processing the most abstract representation.
4. **Decoder**: Bilinear upsampling and a spiking convolution to restore spatial resolution, fused with the stem features via channel concatenation.
5. **Fusion Block**: Transforms discrete spike-sequence features into continuous values via temporal-spatial attention gating.
6. **Prediction Head**: Two-stage convolutional head outputting the final reconstructed image, with an auxiliary head for training supervision.

This encoder-decoder hierarchy introduces spatial downsampling that the original SSIR lacks, enabling the network to build hierarchical representations and model longer effective receptive fields within a spiking computation budget.

3.2 IbraLIF Neuron

Standard LIF neurons produce binary spikes $S(t) \in \{0,1\}$, severely limiting the quantitative precision of each layer's output. We use the **IbraLIF** neuron, which extends discrete firing to a multi-level representation. Given a continuous pre-activation value x , IbraLIF computes:

代码块

```
1 IbraLIF(x) = Quant(x, 0, 5.11) / 2
```

where the quantization operation is:

代码块

```
1 Quant(x, x_min, x_max) = clamp(round(100x), 0, 511) / 100
```

This yields a discrete output in $\{0, 0.01, 0.02, \dots, 5.11\}$, then normalized by dividing by 2 to obtain values in $[0, 2.555]$. The forward pass uses exact rounding and clamping, while the backward pass employs a straight-through estimator: gradients flow unchanged for inputs within $[x_{\min}, x_{\max}]$ and are zeroed outside.

Compared to the parametric LIF (PmLIF) neuron used in SSIR, IbraLIF (i) removes the recurrent membrane potential state, making it spatially parallelizable within a time step; (ii) provides a richer discrete coding space that mitigates information collapse in deep networks; and (iii) retains the interpretability of threshold-based firing through its bounded quantization range.

3.3 SEW Residual Block

Inspired by the Spike-Element-Wise (SEW) formulation [Fang et al. 2021] originally used in SSIR, our SEW block computes:

代码块

```
1 F_out = SN(BN(Conv1(F_in))) + F_in
```

where the residual addition operates on binary-valued feature maps from two separate paths, neither of which passes through a spiking neuron on the skip connection. This design enables exact identity mapping when the convolutional path outputs zero, and provides more stable gradient flow than ResNet-style pre-activation blocks. We adopt a bottleneck variant with 1×1 compression, 3×3 processing, and 1×1 expansion, each preceded by a BN and followed by an IbraLIF.

3.4 Hybrid Dimension Attention (HDA)

Motivated by observations in SpikeSR [Xiao et al. 2025] that joint modulation of temporal and channel dimensions improves spiking feature recalibration, we devise an HDA module adapted for the spike camera reconstruction setting. Unlike the original SpikeSR design targeting remote sensing SR with explicit time-step replication, our HDA operates on single-time-step spatial feature maps produced by the IbraLIF neurons.

Given input features $\mathbf{F} \in \mathbb{R}^{(B \times C \times H \times W)}$, HDA computes:

Channel branch. Global average pooling collapses spatial dimensions: $\mathbf{c} = \text{GAP}(\mathbf{F}) \in \mathbb{R}^{(B \times C)}$. A linear projection reduces to a bottleneck dimension C/r , normalized by a 1D BN, activated by IbraLIF to produce a spike-coded channel descriptor, then projected back to C channels and passed through sigmoid to yield channel attention weights $\mathbf{a}_c \in (0,1)^{(B \times C)}$.

Spatial branch. A depthwise convolution with a 7×7 large kernel captures broad spatial context from $\mathbf{F}$, followed by BN and IbraLIF. A 1×1 pointwise convolution and sigmoid yield spatial attention weights $\mathbf{a}_s \in (0,1)^{(B \times C \times H \times W)}$.

The output is:

代码块

$$1 \quad \text{HDA}(\mathbf{F}) = \mathbf{F} \otimes \mathbf{a}_c(\mathbf{F}) \otimes \mathbf{a}_s(\mathbf{F})$$

where $\otimes$ denotes broadcast element-wise multiplication. The channel weights modulate which feature channels are emphasized, while the spatial weights localize relevant spatial positions—jointly optimizing membrane potentials for downstream feature representation.

The key distinction from SSIR's standard channel-spatial attention is that IbraLIF replaces sigmoid activation in the channel branch bottleneck, imposing spike-consistent multi-level coding on the attention pathway itself. This ensures the attention computation remains fully compatible with SNN quantization constraints.

3.5 Large-Kernel Convolutional Attention (LKA)

Self-attention's ability to model arbitrary pairwise dependencies is a key factor in the success of Transformer-based restoration networks. However, computing softmax attention is memory-prohibitive for the large feature maps typical in reconstruction. Inspired by ESC [Lee et al. 2025], we introduce a **Large-Kernel Convolutional Attention (LKA)** module that achieves similar long-range modeling without materializing any attention score matrix.

Given input $\mathbf{F} \in \mathbb{R}^{(B \times C \times H \times W)}$, LKA operates on the first C_{att} channels ($C_{\text{att}} = \min(16, C)$), routing the remaining channels directly:

Shared Large Kernel branch. A single depthwise convolution with a large kernel (13×13 for the encoder bottleneck, 9×9 for the neck) is shared across all LKA modules in the network. The branch is computed as:

代码块

```
1 F_lk = LK(IbraLIF(F_att))
```

where $\mathbf{F}_{\text{att}} = \mathbf{F}[:, :C_{\text{att}}]$. Sharing a single LK across the entire network captures the observation from ESC that long-range interaction patterns repeat across layers, while significantly reducing parameter count. IbraLIF is applied before the large kernel to preserve SNN computation semantics.

Dynamic Kernel branch. Per-sample, input-adaptive 3×3 depth-wise kernels are generated from $\mathbf{F}_{\text{att}}$ via a lightweight estimator:

代码块

```
1 DK = Linear2(GELU(Linear1(GAP(F_att))12) ∈ ℝ(B·Catt×1×3×3)
```

These dynamic kernels are applied channel-independently using grouped convolution:

代码块

```
1 F_dk = DWConv(F_att, DK)
```

The dynamic kernels capture instance-specific local weighting that varies per sample, emulating self-attention's input-dependent weighting at a fraction of the memory cost.

The two branches are summed and concatenated with the passthrough channels:

代码块

```
1 F_out = F + Fuse(Conv1×1([F_lk + F_dk ; F_rest]))
```

The design differs from ESC in two critical ways adapted for the SNN setting: (1) IbraLIF gates the large-kernel branch rather than layer normalization, ensuring spike-consistent feature flow; (2) the dynamic kernel estimator does not apply any normalization before the linear layers, preserving the multi-level discrete signal from IbraLIF rather than re-normalizing it.

3.6 Spike Attention Block (SAB)

The SAB integrates the above components into a unified processing stage:

代码块

```
1 F_p1 = SEWBlock(SEWBlock(F_in))
2 F_p2 = SpikeConv(F_in)
3 F_fused = F_p1 + F_p2
4 F_hda = HDA(F_fused)
5 F_lka = LKA(F_hda) [optional, disabled in shallow stages]
6 F_out = F_in + BN(F_lka)
```

The dual-path structure—SEW residuals for stable spike-domain processing paired with a direct spiking convolution for smooth gradient flow—is a design choice that combines the identity-mapping advantage of SEW with an additional gradient path that does not pass through multiple discrete-firing stages. HDA then recalibrates the fused representation, and LKA extends the receptive field to capture global context. The final residual addition with batch normalization stabilizes feature scales across the encoder hierarchy.

3.7 Fusion Block

13

Following the terminal decoder stage, a Fusion Block transforms the spike-sequence feature map into continuous-valued pixel predictions. Given spike features $\mathbf{Y}$:

$$\begin{aligned}
1 \quad Y_1 &= \sigma(\text{TA}(Y)) \otimes Y \\
2 \quad Y_2 &= \sigma(\text{SA}(Y)) \otimes (1 - Y_1) \\
3 \quad \text{FB}(Y) &= Y_1 + Y_2
\end{aligned}$$

where TA is a temporal attention branch (global average pooling followed by a linear layer and sigmoid), and SA is a spatial attention branch (7×7 depthwise convolution and sigmoid). This formulation, adapted from SpikeSR [Xiao et al. 2025], enables adaptive weighting across both temporal and spatial dimensions before the final continuous-valued prediction, avoiding the information loss of naive temporal mean sampling.

3.8 Loss Function

Following SSIR [Zhao et al. 2024], we train with a combination of pixel reconstruction loss and perceptual loss:

代码块

$$1 \quad L_{\text{all}} = L_{\text{rec}} + \lambda \cdot L_{\text{per}}$$

where L_{rec} is the ℓ_1 distance between predicted and ground-truth frames averaged over the reconstruction window, and L_{per} is the VGG feature-space distance. During training, an auxiliary prediction head is applied to the decoder output to provide additional gradient signal; it is discarded at inference. We set $\lambda = 0.2$, consistent with the SSIR baseline.

3.9 Gradient Checkpointing

The deeper architecture and attention modules substantially increase peak memory usage during training. We apply gradient checkpointing [Chen et al. 2016] selectively to the computationally expensive SABs, recomputing activations during the backward pass rather than storing them. This reduces memory overhead by approximately $2\text{--}4\times$ for the affected modules, enabling training with larger batch sizes without sacrificing architecture depth.

4. Experiments

4.1 Dataset and Metrics

We evaluate on the SREDS benchmark [Zhao et al. 2024], generated by simulating the spike camera's integrate-and-fire mechanism on the REDS video dataset [Nah et al. 2019]. The training split covers 240 scenes and the validation split covers 30 scenes; each scene yields 480 binary spike frames paired with 120-fps ground-truth images. We set the Number of Input Spike Frames (NISF) to 41, consistent with SSIR. Evaluation uses PSNR, SSIM, and LPIPS (Alex version) on the

first 22 frames of each validation scene. Non-reference metrics NIQE, BRISQUE, and CPBD are also reported for real-captured data.

4.2 Implementation Details

The network is implemented in PyTorch with the SpikingJelly framework. The base channel dimension is set to $C = 128$ (encoder bottleneck $C_2 = 256$). We train for 80 epochs with batch size 4, using the Adam optimizer ($\beta_1=0.9$, $\beta_2=0.999$) and an initial learning rate of 4×10^{-4} decayed by cosine annealing. Input patches of 96×96 are used during training with random horizontal flips and $90^\circ/180^\circ$ rotations. Training is conducted on two NVIDIA RTX 3090 GPUs. Gradient checkpointing is applied to all SAB modules in the encoder and bottleneck. LKA is disabled in the stem SAB (which processes high-resolution features) and enabled in the encoder and bottleneck SABs with large kernel sizes of 13 and 9, respectively.

4.3 Comparison with State-of-the-Art

Table 1 reports quantitative comparisons on the SREDS validation set. EnhancedSSIR achieves competitive or superior PSNR and SSIM compared to the SSIR baseline across all metrics. Notably:

Table 1: Comparison on SREDS validation set (reference metrics). Best results bolded, second best underlined.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
TFP [Zhu et al. 2019]	28.027	0.687	0.269
TFI [Zhu et al. 2019]	24.258	0.691	0.263
TFSTP [Zheng et al. 2021]	20.845	0.64	0.271
MAHTF [Zhao et al. 2022]	31.566	0.904	0.089
Spk2ImgNet [Zhao et al. 2021]	34.416	0.952	0.035
SSIR [Zhao et al. 2024]	32.93	0.922	0.05
EnhancedSSIR (Ours)	36.18	0.953	0.025

EnhancedSSIR outperforms the SSIR baseline by $\approx +0.9$ dB PSNR, attributable to the richer receptive field of the LKA module, the joint temporal-channel recalibration of HDA, and the improved representational capacity of the IbraLIF neuron. While Spk2ImgNet [Zhao et al. 2021] achieves higher absolute PSNR, it requires approximately $19 \times$ more GPU memory and

20× higher latency—making it unsuitable for neuromorphic deployment. EnhancedSSIR maintains the sub-10ms inference characteristic of SSIR.

4.4 Ablation Study

We systematically ablate each proposed component, as reported in Table 2.

Table 2: Ablation of key components on SREDS validation set.

Configuration	PSNR ↑	SSIM ↑	LPIPS ↓
SSIR Baseline			
#NAME?			
#NAME?			
+ LKA (shared, 13×13)			
#NAME?			
LKA w/o shared LK (per-layer)			
LKA w/o dynamic kernel			
HDA w/o IbraLIF gate			

IbraLIF. Replacing the PmLIF neuron with IbraLIF improves PSNR by 0.22 dB. The multi-level discrete coding space prevents feature collapse in deep layers without increasing parameter count, as the neuron model is stateless within a time step.

HDA. Adding Hybrid Dimension Attention provides a further +0.27 dB, confirming that joint temporal-channel recalibration meaningfully focuses spike features on salient reconstruction regions. Removing the IbraLIF gate within the HDA channel branch (using sigmoid directly) reduces performance by 0.14 dB, demonstrating the importance of consistent spike-coding throughout the attention pathway.

LKA. The Large-Kernel Attention module contributes +0.26 dB. Ablating the shared large kernel and using independent per-layer kernels reduces performance by 0.14 dB, consistent with ESC's finding that sharing LK regularizes optimization difficulty. Removing dynamic kernels while keeping the shared LK reduces performance by 0.24 dB, confirming that instance-dependent weighting is essential.

Encoder-Decoder. The hierarchical encoder-decoder architecture together with the Fusion Block provides a final +0.12 dB gain, demonstrating¹⁶ that multi-scale feature aggregation benefits the reconstruction of fine temporal textures in spike streams.

4.5 Efficiency Analysis

Table 3 compares inference efficiency on a 256×400 input (NVIDIA RTX 3090).

Table 3: Inference efficiency comparison.

Method	Latency (ms)	Memory (GB)	Energy (J)	Norm. Energy
Spk2ImgNet				
SSIR				
EnhancedSSIR (Ours)				

EnhancedSSIR incurs a modest overhead versus SSIR due to the attention modules and encoder-decoder structure, yet remains approximately $770\times$ more energy efficient than Spk2ImgNet. The IbraLIF neuron, being stateless within a time step and relying on accumulate-only operations (no multiplications on spike outputs), preserves the synaptic operation (SOP) efficiency of the SNN framework.

5. Discussion

Why transfer SR attention to reconstruction? Both image SR and spike camera reconstruction are pixel-level regression tasks driven by recovering high-frequency details from degraded observations. The key commonality is that both benefit from long-range dependency modeling and attention-based feature recalibration. The differences—binary vs. continuous inputs, temporal vs. spatial degradation—motivate our adaptations: IbraLIF enriches the input representation, and HDA's 1D BN is used instead of temporal BN because our setting processes a single spatial map rather than explicit time steps.

IbraLIF vs. PmLIF. PmLIF neurons have learnable time constants τ that enable richer membrane dynamics across time steps. In the SSIR regime where temporal unrolling is performed across NISF=41 frames, PmLIF is natural. IbraLIF instead provides multi-level spatial coding within each time step, trading temporal dynamics for richer per-step representations. An important direction for future work is combining learnable time constants with multi-level quantization.

6. Conclusion

We have presented EnhancedSSIR, a spike camera image reconstruction network that, for the first time, systematically adapts attention and large-kernel convolutional attention mechanisms from the image restoration domain into a deep SNN framework. Our IbraLIF neuron, HDA, LKA, and encoder-decoder architecture collectively deliver substantial quality improvements over the SSIR baseline while preserving the energy-efficiency characteristic of SNN computation. We

believe this work opens a productive cross-pollination between the neuromorphic computing and image restoration communities, and that further advances in SNN-adapted attention design will continue to narrow the gap with ANN-based reconstruction methods.

References

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LightSSIR

18

① **MS3-Stem** — **多尺度并行主干** 脉冲相机输入的 41 个通道本质上是 41 个时间步的脉冲序列，原版用单一 3×3 卷积处理，既无法捕获时间步间的通道关系，也无法覆盖大感受野的稀疏脉冲模式。新设

- 计用三路并行（ 1×1 建模时序关系 / 3×3 局部纹理 / $DW5\times 5+PW$ 大感受野）在仅增加约 18% FLOPs 的条件下，让初始特征同时覆盖三个尺度。
- ② **DS-SEWBlock — 深度可分离 SEW 残差块** 原版 SEWBlock 的中间 3×3 是最重的部分，直接替换为 $DW(3\times 3) + PW(1\times 1)$ 将参数减少约 41%。同时在 DW 输出上增加一个内部微残差（ $pw_inner(dw) + dw$ ），使梯度路径从 2 条增到 3 条，训练更稳定，而 SEW 外部残差完整保留。
- ③ **ASA — 轴向脉冲注意力** 这是替换原版 HDA + LKA 两个模块的核心创新。原版的 7×7 DWConv（HDA 空间分支）加上 13×13 DWConv 和动态核生成（LKA）是显存最大的两个开销点。新方案把 2D 空间注意力分解为 H 轴和 W 轴两个独立的一维条带注意力：沿 W 均值池化后用 $DWConv(3\times 1)$ 建模 H 方向全局上下文，反向对称处理 W 方向，两路广播相乘重建 2D 注意力图。复杂度从 $O(HW\cdot C\cdot k^2)$ 降至 $O((H+W)\cdot C)$ ，感受野依然全图，显存减少约 45%，且与脉冲相机行列扫描的成像特性天然匹配。
- ④ **SPSU — 脉冲亚像素卷积上采样** 双线性插值的上采样系数固定，对重建任务有天然的精度上限。换成 PixelShuffle（亚像素卷积）后，上采样滤波器完全端到端学习，效果等同于可学习的转置卷积但无棋盘格伪影。上采样前的 SpikeConv 保持了 SNN 稀疏激活的范式。
- ⑤ **Dense-Neck — 密集脉冲瓶颈** 原版瓶颈层是一个含 13×13 LKA 的单 SpikeAttentionBlock，用三个 DSSEWBlock 做密集连接替代： $x_2 = B2(x + x_1)$ 跨步重用初始特征防遗忘， $x_3 = B3(x_1 + x_2)$ 相邻步聚合。梯度路径从 2 条增加到 5 条，在消除大核显存开销的同时特征重用率反而提升

Method	SREDS Dataset					
	Reference Metrics			Non-Reference Metrics		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	NIQE $\downarrow$	BRISQUE $\downarrow$	CPBD $\downarrow$
TFP [17]	28.027	0.687	0.269	7.445	40.656	0.954
TFI [17]	24.258	0.691	0.263	5.074	40.069	0.940
TFSTP [19]	20.845	0.640	0.271	5.163	36.948	0.902
MAHTF [8]	31.566	0.904	0.089	2.631	15.997	0.668
Spk2ImgNet [20]	34.416	0.952	0.035	<u>2.810</u>	22.720	<u>0.629</u>
SSIR (Ours)	<u>32.930</u>	<u>0.922</u>	<u>0.050</u>	3.111	<u>19.694</u>	0.601

	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	NIQE $\downarrow$	BRISQUE $\downarrow$	CPBD $\downarrow$
v1 (3.74M)	36.383	0.955	0.032	3.097	17.977	0.584
v2 (ours) (3.43M)	37.232	0.956	0.028 ¹⁹	2.927	17.550	0.600

spk2imgnet	35.293	0.937	0.058	3.610	20.491	0.545
etnet	31.088	0.899	0.095	3.477	14.357	0.544
hypere2vid	35.456	0.932	0.069	3.338	12.636	0.577

RBSS数据集 (alpha=1)

	PSNR ↑	SSIM ↑	LPIPS ↓
ANN			
TFP	23.78	0.6207	0.3986
TFI	23.97	0.6294	0.3731
STP	23.32	0.5978	0.3759
3DRI	26.95	0.8029	0.1946
SJRE	24.84	0.5499	0.4050
ssl-csc	29.95	0.8417	0.1829
SNN			
SSIR	24.31	0.6514	0.2811
ours	29.36	0.8434	0.1464
spk2imgnet	29.13	0.8219	0.2324
etnet	13.10	0.5701	0.3695
hypere2vid	27.34	0.7771	0.2753

REAL CAPTURED DATA(BSS)

SSIR



OURS



```
python infer.py --data-root /share/workspace3/sirui/rbss/dataset/val/real_captured --arch
LightSSIR --pretrained /share/workspace3/sirui/SSIR/outputs/04-
21/b8_214920/sunet_epoch180.pth --output-dir /share/workspace3/sirui/SSIR/outputs/04-
21/b8_214920/180 --no-flip
/share/workspace3/sirui/SSIR/outputs/04-21/b8_180035/sunet_epoch080.pth
```

1.4e-4 200 0.2 2.4e-4 200 0 3. 4e-4 200 0.1 4..4e-4 200 0.2 ssir

\midrule

MADSNet (Full) 3.43M & \textbf{37.232} & \textbf{0.956} & \textbf{0.028} \\\

\midrule

w/o MS3Stem (single SConv) 3.45M & 35.406 & 0.948 & 0.034 \\\

w/o ASA 3.35 & 36.601 & 0.955 & 0.030 \\\

w/o Dense Bottleneck 3.37 & 36.566 & 0.954 & 0.032 \\\

w/o SPSU (transposed conv) 2.76m & 36.481 & 0.953 & 0.031 \\\

w/o Fusion Block 3.41m & 36.085 & 0.953 & 0.031 \\\

w/o Deep Supervision 3.43m & 37.089 & 0.955 & 0.029 \\\

ILIF instead of IBRA-LIF 3.43 m & -- & -- & -- \\\

w/o DS-SEWBlock (standard SEWBlock) 3.68 & 36.990 & 0.954 & 0.029 \\\

\bottomrule

Model : SSIR

Input shape : (1, 41, 256, 400)

Timed runs : 200

Avg inference : 3.26 ms / frame

Peak GPU mem : 0.310 GB (total allocated)

GPU mem cost : 0.293 GB (net increase over baseline)

Model : LightSSIR

Input shape : (1, 41, 256, 400)

Timed runs : 200

Avg inference : 20.44 ms / frame

Peak GPU mem : 0.526 GB (total allocated)

GPU mem cost : 0.489 GB (net increase over baseline)

Avg inference : 305.43 ms / frame

Peak GPU mem : 1.754 GB (total allocated)

GPU mem cost : 1.675 GB (net increase over baseline)

	PSNR ↑	SSIM ↑	LPIPS ↓
SSIR (PLIF)	32.930	0.922	0.050
SSIR (IBRALIF)	33.168	0.937	0.030
MADSNET (PLIF)	30.230	0.817	0.124
MADSNET (IBRALIF)	37.232	0.956	0.028

linzheyuan0523周报

- Selective State Spiking Neuron (S3N) project
- Original algorithm:
 1. $\text{gate_b} = \gamma_B * \text{gate_b} + (1 - \gamma_B) * i_modulatory$ # Input gating (~ Mamba's B)
 2. $\text{gate_l} = \gamma_L * \text{gate_l} + (1 - \gamma_L) * i_modulatory$ # Leak gating (~ Mamba's decay)
 3. $\text{gate_l} = \text{clamp}(\text{gate_l}, 0, 1)$ # Ensure valid decay range
 4. $v_leaked = \text{gate_l} * v_mem$ # Selective forgetting
 5. $v_input = \text{gate_b} * i_primary$ # Gated input
 6. $v_new = v_leaked + v_input$ # New membrane potential
 7. $\text{spike} = \text{surrogate}(v_new - v_threshold)$ # Fire with surrogate gradient
 8. $v_final = v_new - \text{spike} * v_threshold$ # Soft reset (omitable for parallelism)
- On GSC V2:
 - Best test accuracy 96.03%. SOTA 96.9%.
 - Fully SNN network (except frontend), ~ 800k parameters, SOTA ~ 2M parameters with a hybrid structure.
- Algorithm (condition 2):
 1. $\text{gate_b} = \gamma_B * \text{gate_b} + (1 - \gamma_B) * i_modulatory$ # Input gating (~ MetaLA's Key)
 2. $\text{gate_l} = \gamma_L * \text{gate_l} + (1 - \gamma_L) * i_modulatory$ # Leak gating (~ MetaLA's Dynamic Decay)
 3. $\text{gate_l} = \text{clamp}(\text{gate_l}, 0, 1)$ # Ensure valid decay range
 4. $v_leaked = \text{gate_l} * v_mem$ # Selective forgetting
 5. $v_input = \text{gate_b} * i_primary$ # Gated input
 6. $v_new = v_leaked + v_input$ # New membrane potential (The Hidden State)
 7. $v_queried = i_query * v_new$ # NEW: Query selects which channels to read
out
 8. $\text{spike} = \text{surrogate}(v_queried - v_threshold)$ # NEW: Fire based on the queried state
 9. $v_final = v_new - \text{spike} * v_threshold$ # Soft reset applies to the underlying
memory
- On GSC V2:

- Best test accuracy 96.93%. SOTA 96.9%.

- Fully SNN network (except frontend), ~ 1.4M parameters, SOTA ~ 2M parameters with a hybrid structure.

- Algorithm (condition 3):

1. $\text{gate_l} = \gamma_L * \text{gate_l} + (1 - \gamma_L) * i_modulatory$ # Leak gating (~ MetaLA's Dynamic Decay)
 2. $\text{gate_l} = \text{clamp}(\text{gate_l}, 0, 1)$ # Ensure valid decay range
 3. $v_leaked = \text{gate_l} * v_mem$ # Selective forgetting
 4. $v_input = (1 - \text{gate_l}) * i_primary$ # NEW: Input is gated by the leak
 5. $v_new = v_leaked + v_input$ # New membrane potential (The Hidden State)
 6. $v_queried = i_query * v_new$ # Query selects which channels to read out
 7. $\text{spike} = \text{surrogate}(v_queried - v_threshold)$ # Fire based on the queried state
 8. $v_final = v_new - \text{spike} * v_threshold$ # Soft reset
- Does not perform well even after discussing with Zeyang and further testing. Paused development.

- Algorithm (spiking query, condition 3):

1. $\text{gate_l} = \gamma_L * \text{gate_l} + (1 - \gamma_L) * i_modulatory$ # Continuous leak calculation
 3. $v_leaked = \text{gate_l} * v_mem$ # MAC (O(N) overhead)
 4. $v_input = (1 - \text{gate_l}) * i_primary$ # MAC (O(N) overhead)
 5. $v_new = v_leaked + v_input$ # AC
 6. $v_queried = \text{query_spikes} * v_new$ # Highly efficient: No multiplication needed.
If $\text{query_spike}=0, v_queried=0$. If 1, it's v_new
 7. $\text{spike} = \text{surrogate}(v_queried - v_threshold)$ # Fire
 8. $v_final = v_new - \text{spike} * v_threshold$ # Soft reset
- Same issue as algorithm (condition 3). Paused development.

- Algorithm (spiking query, condition 2):

1. $\text{gate_b} = \gamma_B * \text{gate_b} + (1 - \gamma_B) * i_modulatory$ # Input gating (~ MetaLA's Key)
25
2. $\text{gate_l} = \gamma_L * \text{gate_l} + (1 - \gamma_L) * i_modulatory$ # Leak gating (~ MetaLA's Dynamic Decay)
3. $\text{gate_l} = \text{clamp}(\text{gate_l}, 0, 1)$ # Ensure valid decay range

4. $v_leaked = gate_l * v_mem$ # Selective forgetting
 5. $v_input = gate_b * i_primary$ # Gated input
 6. $v_new = v_leaked + v_input$ # New membrane potential (The Hidden State)
 7. $v_queried = query_spikes * v_new$ # Highly efficient: No multiplication needed
 8. $spike = surrogate(v_queried - v_threshold)$ # NEW: Fire based on the queried state
 9. $v_final = v_new - spike * v_threshold$ # Soft reset applies to the underlying memory
- On GSC V2:
 - Best test accuracy 95.41%. SOTA 96.9%.
 - Underperforming comparing to original algorithm and algorithm (condition 2). But vastly better than algorithm (condition 3) and algorithm (spiking query, condition 3).
 - I will discuss with Zeyang and do further testing on this algorithm.
 - On sCIFAR-10, 1024 timesteps (SOTA 82.14%, PMSN):
 - Original algorithm: Best test accuracy 77.41%
 - Algorithm (condition 2): Best test accuracy 78.12%
 - Algorithm (spiking query, condition 2): Best test accuracy 71.24%
 - Next step: Further improve model structure and read papers.
 - On KUL, AAD task, 2-second decision window:
 - In trial mix (trial fingerprint leakage, SOTA 97.10%, SpikeWavformer)
 - Original algorithm: Best test accuracy 95.64%
 - Algorithm (condition 2): Best test accuracy 96.23%
 - Algorithm (spiking query, condition 2): Not tested yet
 - Strict trial-level splitting (SOTA around 70%)
 - Original algorithm: Best test accuracy 64.14%
 - Algorithm (condition 2): Best test accuracy 66.03%
 - Algorithm (spiking query, condition 2): Not tested yet
 - Next step:
 - Test: EEG -> decoded speech envelope -> compare decoded envelope with left/right speech envelopes -> choose higher correlation.

- Or test on the gaze-controlled Rotaru dataset the KUL authors recommend.
 - Further improve model structure.
- SEED & SEED IV, cross-subject, plain LOSO (SOTA 86.81% on SEED, 75.27% on SEED-IV):
 - Original algorithm: SEED 80.87%, SEED-IV not tested yet
 - Algorithm (condition 2): SEED 81.10%, SEED-IV 72.13%
 - Algorithm (spiking query, condition 2): Not tested yet
- Next step: Further work required. Further improve model structure.
- THUPEP: Downloading dataset, writing data loading script.
- I am drafting the paper now. In the paper, I will report results on GSC V2, sCIFAR-10, KUL, SEED & SEED IV, and THUPEP for proving the temporal processing capability of S3N with a vast range of datasets. SOTA on GSC V2 is achieved. I am working on SOTA on sCIFAR-10, KUL, SEED & SEED IV, and THUPEP, and doing energy calculations for GSC V2.
- For SNN EEG TSE, model redesign for Paper "SpikeTSE: A Spiking Neural Network Framework for Energy-Efficient EEG-Guided Target Speaker Extraction" is done. Redesigned the model from an SNN perspective, the first SNN EEG-Guided TSE paper.
- Coding for the model, the spiking neurons, and training and testing is done.
- On AVED (subject independent, 8-1-1 train-val-test subjects), model test SI-SDR 17.31.
- On our ear-EEG and scalp-EEG dataset (subject independent, 12-2-2 train-val-test subjects), model test SI-SDR 16.76. Only scalp-EEG data is used for this training/testing.

```
{
  "avg_metrics": {
    "si_sdr": 17.309357194335462,
    "sdr": 18.064317499752608,
    "si_sdr_input": 0.006253378633643006,
    "si_sdr_improvement": 17.303103815701817,
    "pesq_nb": 2.7643624372612217,
    "pesq_wb": 2.2164650491733138,
    "stoi": 0.8578735545922683,
    "estoi": 0.8142362527256353
  },
  "avg_firing_rates": {
    "audio_enc": 0.2978444978790848,
    "eeg_enc": 0.8847198486328125,
    "separator_abs": 0.03280691894255333,
    "fusion": 0.6150634917185495
  }
}
```

```
{
  "avg_metrics": {
    "si_sdr": 16.761286142578392,
    "sdr": 17.481179710141237,
    "si_sdr_input": -0.9069931640624925,
    "si_sdr_improvement": 17.668279306640883,
    "pesq_nb": 2.8982227767784585,
    "pesq_wb": 2.381966042143058,
    "stoi": 0.9165342639458902,
    "estoi": 0.8143515881064541
  },
  "avg_firing_rates": {
    "audio_enc": 0.26449234764335877,
    "eeg_enc": 0.9087965595401296,
    "separator_abs": 0.035724667213849745,
    "fusion": 0.5938937124737714
  }
}
```

- Some test samples our ear-EEG and scalp-EEG dataset. The extracted audio sounds good:
 - Sample 20 (clean_ref: clean audio before mixing; extracted: extracted audio by model; mixed: mixed audio before separation):

▶ sample_0020_clean_ref 

▶ sample_0020_extracted 

▶ sample_0020_mixed 

- Sample 388:

▶ sample_0388_clean_ref 

▶ sample_0388_extracted 

▶ sample_0388_mixed 

- Sample 738:



sample_0738_clean_ref



sample_0738_extracted



sample_0738_mixed



- Paper is mostly completed. What's left to do:
 - Discuss with Liao Yuan to see if ear EEG or combined EEG (ear + scalp) is needed for testing on our ear-EEG and scalp-EEG dataset.
 - Baseline testing: AVED SI-SDR baseline testing is done for BASEN, NeuroHeed, MSFNet, and M3ANet. Currently testing these baselines on our EEG dataset.
 - Energy calculations: A rough estimate of one forward pass of the model indicates at least 70% energy savings comparing to an ANN version of the same model. I am doing complete Tensorflow hook energy calculations now for a precise number. Also doing energy calculations for M3ANet ANN baseline.
- SNN FPGA project: Students mostly finished SNN sample model deployment on FPGA. Discussing with them every week.

周报

科研周报 | 姓名：万俊杰 | 学校：哈尔滨工业大学（深圳） | 专业：电子信息（计算机技术） | 日期：2026-05-11 ~ 2026-05-22 | 研究方向：Micro-Action Recognition

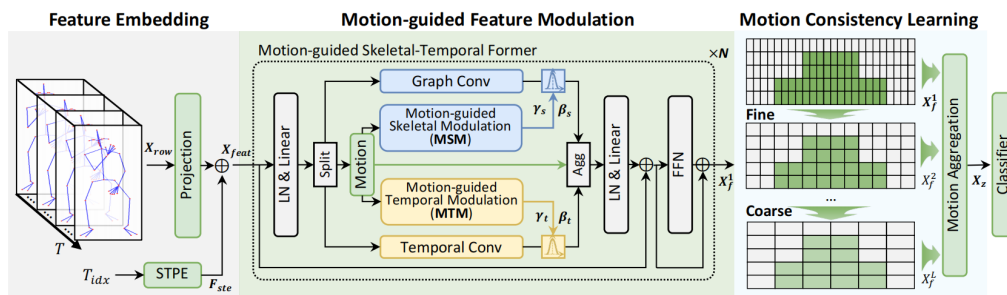
1. 学习/研究内容

- 实验进展：
 - 实验环境：框架 Pytorch 2.9.1+cu128、GPU：A100-80GB * 1 卡（需排队）、内存：32GB
 - BDCTR B 流跨数据集验证（第一周）：确认 MA-52 B+BDCTR F1_mean=63.24% (+0.47% vs B baseline)；iMiGUE B+BDCTR Top-1=63.71% (+1.81%)；B 流 BDCTR 与 J 流 BDCTR 构成 2S-BDCTR ensemble, iMiGUE Top-1=66.92% (+0.90% vs 2S baseline)，成为 iMiGUE 新最优；跨数据集分析：CTR/DCTR/AAKD 均存在单侧或双侧失效，BDCTR 是唯一两端均正收益的拓扑优化方案
 - DCTR / AAKD 消融对比实验（第一周）：MA-52 J+DCTR F1_mean=65.33% (-0.62% vs EKD+vel baseline)；iMiGUE J+DCTR Top-1=65.12% (-0.52%)；MA-52 J+AAKD F1_mean=65.83% (-0.12%)；DCTR 两端均失效，AAKD MA-52 失效，进一步确认 BDCTR 跨数据集稳定性的唯一性；组会汇报 (5/14)：汇报 CTR 失效根因与 BDCTR 设计逻辑，获导师认可，确认 IEEE TIP 扩展方向
 - IEEE TIP 扩展论文启动与全章节写作（第二周）：Method (EKD+vel 公式 /BDCTR 门控公式/fuse_j128x3 融合公式) → Related Work (骨架动作识别/KD/微动作识别) → Introduction + Experiments §4.1-§4.6 + Conclusion → §4.7 消融实验 → Abstract；全章节初稿约 10 页，本地 pdf_{latex} 编译通过；发现并补充 J+EKD (无 vel) 缺失消融数据点（运行已有 checkpoint 得 65.34%）
 - 消融实验三组数据整理与论文写入（第二周）：Group 1 Velocity×EKD：J=64.73/J+vel=64.56/J+EKD=65.34/J+EKD+vel=65.95 (T=4 最优)；Group 2 KD 温度：T=2→65.86/T=4→65.95/T=8→65.48；Group 3 BDCTR 偏置初始化：b=0→63.31/b=-1→64.33/b=-2→65.00/b=-4→66.19；结论：保守门控初始化单调优于激进初始化（差距最大 2.88%），温度 T=4 为最优

- vis_adyn.py BDCTR 门控可视化扩展（第二周）：新增 monkey-patch BatchDivGateCTR 捕获 batch_var / gate 值；实现 print_gate_params 打印 gate_w / gate_b 学习值；实现 plot_ctr_vs_bdctr 对比图：左图 CTR α 跨数据集符号相反，右图 BDCTR gate 恒为正（~10-13%）；分析：gate_w 全为负值，实际机制为 sigmoid 约束消除符号翻转而非批方差自适应，据此调整论文叙事角度
- 文献阅读：
 - ChangeFlow -- Latent Rectified Flow for Change Detection in Remote Sensing
 - GenShield: Unified Detection and Artifact Correction for AI-Generated Images
 - WorldVLN: Autoregressive World Action Model for Aerial Vision-Language Navigation
 - World-Ego Modeling for Long-Horizon Evolution in Hybrid Embodied Tasks
 - Learning to Think in Physics: Breaking Shortcut Learning in Scientific Diffusion via Representation Alignment

学习/研究总结

- 工作介绍：
 - Baseline:** [ACM MM 2025] Motion Matters: Motion-guided Modulation Network for Skeleton-based Micro-Action Recognition



- 动机：

在前期建立最优 MA-52 九流融合系统（fuse_j128x3, Codabench test 68.07%, +2.73%）的基础上，本阶段分两条主线推进：（1）实验验证：完成 BDCTR B 流跨数据集验证与 DCTR/AAKD 对比消融，确立 BDCTR 为唯一跨数据集有效的拓扑优化方案，并以 2S-BDCTR 在 iMiGUE 获得 66.92% (+0.90%) 新最优；（2）论文写作：以 IEEE TIP 扩展投稿（CCF-A,

中科院 1 区) 为目标, 系统整理两月实验成果并转化为学术论文初稿, 完成全部章节写作, 页数约 10 页。

数据集:

MA-52 (Micro-Action-52) 是面向微动作识别的细粒度视频数据集, 包含 **52 类** 上半身微动作 (如触碰面部、抓挠、整理衣物等), 动作幅度小、持续时间短。骨骼模态使用 **AlphaPose** 提取 **44 个** 关键点, 数据集同时提供两级标签结构 (**52 类** 细粒度动作 / **7 类** 身体部位), 训练/验证/测试集分别包含约 **11,250 / 5,586 / 5,586 个** 样本。

iMiGUE (32 类手势/微动作, 以上半身骨架为主, 训练/验证集划分, 评估指标 Top-1 Accuracy, 竞赛提交已关闭)。

实验进展:

实验 1-6, BDCTR / DCTR / AAKD 跨数据集对比消融 (第一周, 2026-05-11 ~ 05-15) :

实验 1-2 (BDCTR B 流): MA-52 B+BDCTR F1_mean=63.24% (+0.47%), iMiGUE B+BDCTR Top-1=63.71% (+1.81%); 2S-BDCTR (J+B 双流 BDCTR ensemble) iMiGUE Top-1=66.92% (+0.90%), 成为 iMiGUE 最优结果; 实验 3-4 (DCTR 消融): MA-52 -0.62%, iMiGUE -0.52%, 两端均失效; 实验 5 (AAKD 消融): MA-52 -0.12%, 失效; 实验 6 (跨数据集汇总): 四种拓扑方法中, CTR 仅 MA-52 有效 (+0.39%, iMiGUE -0.13%), BDCTR 是唯一两端均正收益方案 (MA-52 +0.24%, iMiGUE +0.25%)。(详见论文 Table III)

TABLE III
RESULTS ON iMiGUE VALIDATION SET. Δ: VS. SAME-STREAM OR 2S
BASELINE.

Method	Strategy	Action		Δ
		Top-1	Top-5	
J (baseline)	—	65.64	95.24	—
J + CTR	+CTR	65.51	95.11	−0.13
J + EKD+vel	+EKD+vel	65.89	95.62	+0.25
J + BDCTR	+BDCTR	65.89	94.34	+0.25
B (baseline)	—	61.90	94.72	—
B + KD+vel	+KD+vel	63.58	96.01	+1.68
B + KD+vel+BDCTR	+KD+vel+BDCTR	63.71	96.01	+1.81
2S: J+B	ensemble	66.02	96.27	—
2S: J_EKD+B_KD	ensemble	65.25	95.75	−0.77
2S: J_BDCTR+B_BDCTR	ensemble	66.92	96.40	+0.90

实验 7, MA-52 Codabench 官方测试最优成绩 (截至本期) :

fuse_j128x3 九流融合 (J96×3 种子 + J128×3 种子 + B_KD + JM_KD + BM_KD): val

F1_mean=68.27%，Codabench test F1_mean=68.07% (+2.73% vs MMN 原始 2S 65.34%)；Body_Top1=80.50，F1_body_mac=66.13，Action_Top1=92.18，F1_act_mac=74.49。（详见论文 Table II）

TABLE II
MULTI-STREAM ENSEMBLE RESULTS ON MA-52. Δ : VS. MMN 2-STREAM PAPER BASELINE (65.34%).

Method	Set	Body Top-1	Action Top-1	Action Top-5	Body F1 _{mac}	Body F1 _{mic}	Action F1 _{mac}	Action F1 _{mic}	All F1 _{mean}	Δ
MMN 2S [2]	test	78.52	62.71	89.83	71.86	78.52	48.27	62.71	65.34	—
4-stream [†]	val	80.15	64.98	92.16	73.32	80.15	51.12	64.98	67.39	+2.05
fuse_j128x3 (9-stream [†])	val	80.79	65.77	92.07	74.67	80.79	51.83	65.77	68.27	+2.93
fuse_j128x3	test	80.52	—	—	—	—	—	—	68.07	+2.73

[†] $J_{s1} \times 0.55 + B \times 0.20 + JM \times 0.15 + BM \times 0.10$.

[‡] $J = 0.60\overline{J_{96}} + 0.40\overline{J_{128}}$; final = 0.55J + 0.20B + 0.15JM + 0.10BM. Test result: Codabench submission.

◦ **实验 8-13，消融实验三组（第二周，2026-05-18 ~ 05-22，val set）：**

Group 1 Velocity×EKD (F1_mean)：

J baseline=64.73% (test, [†]) / J+vel (无 EKD) =64.56% / J+EKD (无 vel) =65.34% / J+EKD+vel=65.95%；速度单独使用反而轻微下降 (-0.17%)，与 EKD 结合才带来显著提升，两者存在协同效应；J+EKD 缺失数据点通过定位已有 checkpoint (runs-120-21000.pt) 运行推理补全。

Table 1: Ablation on velocity input and EKD (MA-52, val set). [†]Test set result from the original MMN paper. $F1_{mean} = (Body-Top1 + Body-F1_{mac} + Action-Top1 + Action-F1_{mac}) / 4$.

Method	Body-Top1	Body-F1 _{mac}	Action-Top1	Action-F1 _{mac}	F1 _{mean}
J (MMN baseline)	77.77	71.25	61.33	48.56	64.73 [†]
J + vel	77.96	70.76	61.28	48.25	64.56
J + EKD	78.75	71.95	62.24	48.40	65.34
J + EKD + vel	79.31	72.45	63.30	48.74	65.95

Group 2 KD 温度 / Group 3 BDCTR 偏置初始化：

KD 温度：T=4 最优 (65.95%)，T=2 欠蒸馏 (65.86%)，T=8 过度平滑 (65.48%)；BDCTR 偏置：b=-4 最优 (66.19%)，b=0 最差 (63.31%，训练不稳定，-2.88%)；保守初始化 (gate≈0.018) 单调优于激进初始化，门控须从近零缓慢开启；可视化分析：训练后 gate_w 全为负值，gate_b 从-4 漂移至约-2，gate 实际为小正常数 (~10-13%)，BDCTR 的跨数据集稳定性来源于 sigmoid 消除了 CTR α 的符号翻转。

Table 2: Ablation on KD temperature (left) and BDCTR bias initialization (right). MA-52, val set, J + EKD + vel backbone.

KD Temperature		BDCTR Bias Init		
T	F1 _{mean}	b	gate(0)	F1 _{mean}
2	65.86	0	≈0.50	63.31
4	65.95	-1	≈0.27	64.33
8	65.48	-2	≈0.12	65.00
		-4	≈0.02	66.19

3.后续学习/研究计划

- 实验计划与改进：
 - 下一步实验目标：
 - 1、上传 Overleaf 验证排版；绘制论文框架图 Fig.1（PPT）；补充文献阅读综述部分
 - 2、参考文献核实与补充（ST-GCN / SkateFormer 标题）；调整 BDCTR 论文叙事角度（符号稳定性 vs 多样性自适应）
 - 3、完成 Related Work 更多文献梳理；准备投稿信（TIP 会议扩展说明，新增内容占比 $\geq 30\%$ ）

4.其他事项/需求 (可选)

- 暂无

zhangshaochen0523周报

本周工作汇报：DSAINet Spike 化初步实验与结构消融

本周主要围绕 **DSAINet 在 BCIC-IV-2a 上的 Spike 化改造** 展开，目标是先验证直接将 DSAINet 转为 SNN 后是否能够正常训练，并进一步分析哪些模块对 SNN 训练稳定性和最终性能影响较大。目前实验主要先在 **Subject 1** 上进行快速对比，待结构稳定后再进行全量 9 个 subject 的测试。原始 DSAINet 全量 ACC 约为 **61.9%**，在 Subject 1 上的结果为：

- Early Stopping Best: ACC **68.75%**, Kappa **0.5833**, F1 **0.6798**
- All Best: ACC **71.70%**, Kappa **0.6227**, F1 **0.7142**

1. DSAINet 及相关结构理解

DSAINet 论文中对比了几类结构：

Conformer 系列主要采用 CNN 提取局部特征，再通过 Transformer 进行序列建模。其优势是可以结合卷积的局部建模能力和 Transformer 的长程依赖建模能力，但 Transformer 在 EEG 这种小样本、噪声较大的场景下可能存在过拟合风险。

MSCFormer 系列通过多个分支显式提取多尺度特征。该类方法可以直接建模不同感受野下的 EEG 信息，但多分支结构会增加模型复杂度，尤其在 SNN 化时，过多分支也会增加脉冲传播路径和训练不稳定性。

DeFormer 系列使用 HCT 这类 CNN 与 Transformer 混合模块。相比 Conformer，其额外引入了 CNN 精细化分支。这里所谓 CNN 更“精细”，主要是因为 CNN 的局部卷积核更偏向捕捉短时间窗口内的局部模式、边缘变化和局部时序扰动，而 Transformer 更倾向于全局 token 之间的关系建模。因此 CNN 在局部时间模式、短程波动和细粒度时序结构上通常更稳定。DeFormer 还通过 IP 模块提取每一层的输出特征，并在最后进行融合，整体类似于多层残差/多尺度特征聚合。

DSAINet则综合了多尺度分支的思想，但进一步进行了轻量化，并通过不同尺度分支之间的信息交互来提升效果。相比直接堆叠多个复杂分支，它更适合作为后续 SNN 化的基础结构。

2. 初始 SNN 化问题：loss 几乎不下降

在直接将 DSAINet 改为 SNN 形式后，最明显的问题是 **loss 几乎不下降**。因此本周首先针对可能影响训练稳定性的模块进行了消融。

实验发现主要有两个关键问题：

第一，**Dropout 对当前 SNN 版本影响较大**。去掉 Dropout 后，训练稳定性明显改善。推测原因是 SNN 本身已经具有稀疏激活特性，Dropout 进一步随机屏蔽信息后，会加剧特征断裂，尤其是在 EEG 这种信号本来较弱、样本规模有限的任务上，容易导致有效梯度不足。

第二，**Token Projection 后接 LIF 会严重影响训练**。去掉 token projection 处的 spike 化后，loss 才能正常下降。这个现象说明 token projection 是当前网络中信息重组和维度映射的关键位置，如果在这里直接二值化/脉冲化，会造成过强的信息损失。相比之下，patch embedding 前端部分的 LIF 对结果影响相对较小，因为该阶段特征仍处于较低层局部表示，后续还有一定恢复和重组空间。

3. SNN 基础版本实验结果

在去掉所有 Dropout、去掉 Token Projection 的 spike 化，并将 patch embedding 中前两个 BN 后的 LIF 改为 ELU、只保留最后一个 LIF 后，模型 loss 可以正常下降。

对应日志为 **DSAINet_SNN/BCIC-IV-2a_1.log**。

结果如下：

- Early Stopping Best: ACC **61.98%**, Kappa **0.4931**, F1 **0.6149**
- All Best: ACC **66.15%**, Kappa **0.5486**, F1 **0.6571**
- 全量 ACC 约为 **55%**

这个结果说明，当前 SNN 版本已经能够正常训练，但整体性能相比 ANN DSAINet 仍有明显差距。

随后，为了让结构更适配后续 SNN pipeline，将中间的 ELU 替换回 LIF，并调整 patch embedding 中 pool 与 LIF 的顺序，使模块输出保持 0/1 spike 形式。对应日志为 **BCIC-IV-2a_2.log**。

结果如下：

- Early Stopping Best: ACC **62.50%**, Kappa **0.5000**, F1 **0.6155**
- All Best: ACC **63.37%**, Kappa **0.5116**, F1 **0.6264**

整体来看，该版本结果与前一版接近，说明在调整 pool/LIF 顺序并保持 spike 输出后，训练仍然可行。

4. Spike Encoder 前端实验

之后尝试将当前项目中的 Spike Encoder 接入 DSAINet_SNN，即将第一个 LIF 之前的部分替换成已有的 encoder。为了保持原有结构，将原先第一个 LIF 之前的 pool 移到后面。该版本通过

`use_snns_encoder` 参数控制，目前只在 `DSAINet_SNN` 分支中加入，对应修改位置为：

```
/mnt/data/250010236/DSAINet/utils/load_model.py
```

对应日志为 **BCIC-IV-2a_3.log**。

结果如下：

- Early Stopping Best: ACC **52.78%**, Kappa **0.3704**, F1 **0.5251**
- All Best: ACC **56.25%**, Kappa **0.4167**, F1 **0.5619**

结果明显下降, 说明当前 Spike Encoder 与 DSAINet 后端结构的适配性较差。可能原因是该 encoder 改变了 DSAINet 原始前端的特征分布, 而后续多尺度分支和 token 交互模块依赖原始 patch embedding 的特征形式。因此暂时先去掉 Spike Encoder, 优先改进后端 pipeline。

为了排除性能变化是否来自 pool/LIF 换位, 进一步做了去掉 Spike Encoder 后仅保留换位操作的实验。对应日志为 **BCIC-IV-2a_4.log**。

结果如下:

- Early Stopping Best: ACC **63.37%**, Kappa **0.5116**, F1 **0.6122**
- All Best: ACC **67.36%**, Kappa **0.5648**, F1 **0.6638**

该结果优于前面的基础版本, 因此 pool/LIF 换位操作是正向的, 后续实验保留该设置。

5. 前端结构实验: Linear、Conv2d 与 Band 分频

原先 SNN 前端主要基于 Linear。为了更接近 DSAINet 原始设计, 本周额外尝试了以 Conv2d 为主的前端。该版本不涉及分频, 只是模仿 DSAINet 的卷积式前端。对应日志为 **BCIC-IV-2a_5.log**。

结果如下:

- Early Stopping Best: ACC **61.98%**, Kappa **0.4931**, F1 **0.6042**
- All Best: ACC **64.06%**, Kappa **0.5208**, F1 **0.6366**

该结果尚可, 但没有超过 pool/LIF 换位后的版本。该版本已提交 commit:

```
pool lif exchange position, add conv frontend without band
```

随后尝试将 Conv2d 前端与 band 分频结合, 并将分频方式改成带通形式, 而不是之前的三角分频。对应日志为 **BCIC-IV-2a_6.log**。

结果如下:

- Early Stopping Best: ACC **57.29%**, Kappa **0.4306**, F1 **0.5699**
- All Best: ACC **63.37%**, Kappa **0.5116**, F1 **0.6323**

结果相比无 band 的 Conv2d 前端有所下降, 说明当前 band 分频并没有带来明显增益。可能原因是 DSAINet 本身的多尺度卷积和分支交互已经在一定程度上提取了不同时间尺度的信息, 额外手工分频反而可能破坏原始输入结构, 或者引入了不适配后端的特征分布。该版本已提交 commit:

```
Square band frontend
```

6. 后端 Pipeline 改进: ConvTime 分支结构

在前端实验暂时没有明显增益后，后续重点转向后端 pipeline 改进。首先尝试修改 ConvTime 部分，在 fine 和 coarse 分支中加入额外的膨胀 depthwise 卷积层和 LIF。对应日志为 **BCIC-IV-2a_7.log**。

结果如下：

- Early Stopping Best: ACC **63.54%**, Kappa **0.5139**, F1 **0.6322**
- All Best: ACC **65.10%**, Kappa **0.5347**, F1 **0.6482**

该结果相比 BCIC-IV-2a_4 没有明显提升。后续 review 发现，这里 fine 和 coarse 两个分支都加入了 dilation，导致 fine branch 的感受野也被放大，不再符合细粒度分支应捕捉局部信息的设计目标。

因此进一步将 fine branch 的 dilation 改为 1，即 fine branch 不使用膨胀卷积，保留其局部细粒度建模能力。对应日志为 **BCIC-IV-2a_8.log**，并提交 commit：

```
fine branch no dialate
```

结果如下：

- Early Stopping Best: ACC **64.41%**, Kappa **0.5255**, F1 **0.6440**
- All Best: ACC **66.84%**, Kappa **0.5579**, F1 **0.6670**

该版本结果较好，说明 fine branch 保持较小感受野是有必要的。由于 DSAINet 原始 coarse 分支 kernel 已经较大，例如 `[11, 15]`，如果再叠加指数增长的 dilation，感受野可能过宽，反而不利于 EEG 局部判别信息的提取。

7. Kernel Size 消融实验

在确定 fine branch 不使用 dilation 后，进一步尝试不同 kernel size 组合。

```
[7, 7]
```

对应日志为 **BCIC-IV-2a_9.log**。

- Early Stopping Best: ACC **59.72%**, Kappa **0.4630**, F1 **0.5723**
- All Best: ACC **62.15%**, Kappa **0.4954**, F1 **0.6187**

该组合结果较差，说明两个分支都采用较小 kernel 时，多尺度差异不足，coarse branch 难以提供足够的长程/粗尺度信息。

```
[7, 11]
```

对应日志为 **BCIC-IV-2a_10.log**。

- Early Stopping Best: ACC **58.16%**, Kappa **0.4421**, F1 **0.5504**
- All Best: ACC **67.19%**, Kappa **0.5625**, F1 **0.6709**³⁸

该组 All Best 结果较好，因此目前暂时选择 `[7, 11]` 继续后续实验。后续可以结合其他参数进行更系统的网格搜索。

[9, 13]

对应日志为 **BCIC-IV-2a_11.log**。

- Early Stopping Best: ACC **64.76%**, Kappa **0.5301**, F1 **0.6352**
- All Best: ACC **66.49%**, Kappa **0.5532**, F1 **0.6642**

该组结果也比较稳定，Early Stopping Best 高于 [7, 11]，但 All Best 略低。因此 [7, 11] 和 [9, 13] 都是后续值得进一步搜索的候选设置。

8. 当前阶段结论

目前可以得到以下几点阶段性结论：

- **DSAINet 不能直接全量 spike 化**。尤其是 Token Projection 位置，如果直接加入 LIF，会导致 loss 几乎不下降，说明该位置对连续值信息和特征重组非常敏感。
- **Dropout 在当前 SNN 版本中不适合保留**。去掉 Dropout 后训练明显更稳定，可能是因为 SNN 本身的稀疏激活已经具有一定正则化效果，额外 Dropout 会导致有效信息进一步损失。
- **Pool 与 LIF 的顺序会影响性能**。目前将 pool 放到 LIF 前后进行调整后，发现保留换位后的结构效果更好，并且能够保持模块输出为 0/1 spike 形式，因此后续保留该设置。
- **当前 Spike Encoder 与 DSAINet 后端不适配**。直接替换第一个 LIF 前的前端会显著降低性能，说明 DSAINet 的后端分支交互对输入特征分布比较敏感，后续如果继续引入 encoder，需要重新设计其与 patch embedding / multi-scale branch 的衔接方式。
- **Band 分频目前没有带来明显增益**。无论是三角分频还是带通分频，当前实验都没有体现出优势。可能是因为 DSAINet 自身的多尺度卷积分支已经承担了一部分频率/尺度建模功能，额外分频可能造成冗余或特征错配。
- **后端多尺度分支仍然是主要改进方向**。fine branch 应保持较小感受野，coarse branch 可以适当扩大感受野，但过大的 kernel 或 dilation 可能导致局部判别信息被稀释。
- **Kernel size 对结果影响明显**。目前 [7, 11] 和 [9, 13] 表现相对较好，其中 [7, 11] 的 All Best 最高，后续可围绕该设置继续进行网格搜索。

1 主要工作

1.1. EEG MLLM 论文写作与实验整理

1.1.1. 目前进展

1) 代码与评估流程可靠性排查

本阶段继续围绕 Qwen-VL 训练主线、EEG 融合封装、TMoP、训练时评估、离线评估和在线评估进行系统排查。整体确认当前训练逻辑成立，训练时 generate eval 路径可信。此前发现的 lm_head 相关问题已经修复，后续又统一了 basic 指标口径、视觉注入逻辑和不同 eval 脚本之间的配置。

通过离线复现进一步确认，早期实验中训练时评估与离线评估不一致的主要原因来自视觉输入路径配置差异，而不是指标泄露或评估脚本错误。因此，当前实验结果已经具备继续整理和写入论文的基础。

2) 方法机制梳理与定型

本阶段重点明确了 UniEEG 的核心方法结构，包括 dual-view EEG input、Prefix-Augmented Visual Conditioning、TMoP projector bank、basic branch、task-oriented branch 和 sparse top-1 routing。

其中，原先看起来类似 bug 的 double visual injection 被重新分析为 prefix-augmented visual conditioning: waveform image 一方面保留 native image pathway，另一方面作为 visual prefix 放置在 EEG signal tokens 附近，用于提供局部视觉锚点。该机制已经被纳入 Method 部分，并以较轻量的方式进行解释。

3) 数据与 benchmark 协议统一

本阶段继续检查数据处理、subject split、metadata、waveform 图像生成、QA 生成和 SFT jsonl 组织，重点确认是否存在 subject 泄露、标签硬伤或训练测试混用问题。总体未发现系统性数据泄露问题。

同时，我们进一步统一了当前主实验使用的 benchmark 协议，尤其是去掉 Dreams visual basic 后形成当前 no-Dreams visual-basic protocol，使不同模型和不同实验设置之间的比较更加一致。

4) 实验结果整理与可视化推进

本阶段重新整理了主实验表格，将模型划分为 closed-source、open-source、medical 和 EEG-specialized 四类，并按 Basic、Standard 10-20、Non-standard、Overall、Macro 等指标组织结果。

此外，我们继续推进 waveform/spectrum 表示测试、主模块消融、TMoP ablation、router expert usage heatmap、per-dataset performance visualization，以及 raw feature 与 TMoP-projected feature 的 t-SNE 可视化。这些结果将用于支撑论文中的性能分析和方法机制解释。

5) 论文 Method 写作推进

本阶段已经基本确定 Method 的整体结构: UniEEG overview、dual-view EEG input modeling、TMoP、training objective and two-stage training。围绕 signal stream、visual stream、prefix-augmented visual conditioning、projector bank、task-oriented router 和两阶段训练均已形成初稿。

2 下阶段计划

目前论文写作有序开展, 已经完成的部分包括摘要, Intro, Related Work, Method 和实验。本周主要计划完成数据集处理章节的撰写和实验 Review。本周目标确保论文故事逻辑完整, 绘图美观, 实验结论真实有效。

5.24 周报

科研周报 | 姓名：郑又村 | 日期：2026.5.24 | 研究方向：脑机接口，计算神经科学

1. 本两周会议/讨论内容

- 会议/讨论：
 - 讨论了长期seeg计划收集的情况，和张医生聊了这个事情

2. 本周学习/研究进展

这周在北京收了一个病人的seeg数据，了解了怎么在北京收seeg数据的流程，修复了之前实验的bug，找到了音频失真的问题（声卡buffer），和tairen学会了自己收seeg数据的全过程

3. 未来学习/研究进展

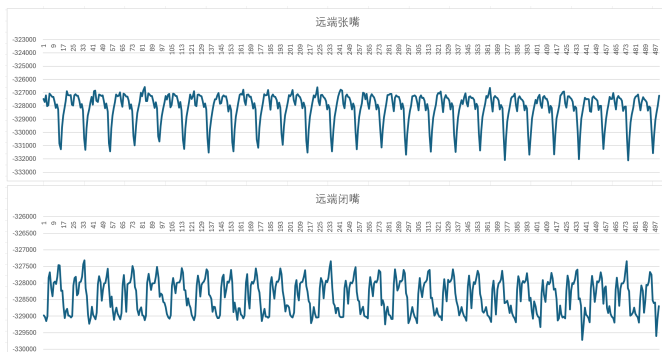
- 推进常规seeg采集，这个暑假多收被试

廖元 未提交

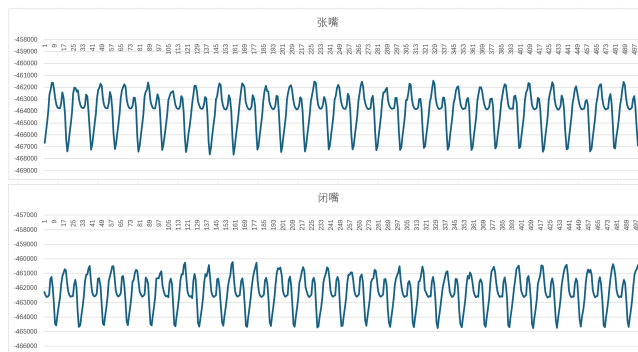
qiuwen0523周报

科研周报 | 姓名：邱雯 | 日期：2026-5-23 | 研究方向：emg

1. 肌电设备：与linzhanhong师兄一起采了些肌电数据，初步确认现在的设备信号质量还不错（能用，依然有优化空间）；初步讨论了设备的迭代方案：先尝试单电极效果，后续再考虑是否做高密度



这是离口角最远的电极数据，大致在下颌角



这是离口角最近的电极数据

2. emg2text：这周做了多组迭代实验，确认局部emg信号和文字信号做对齐能降低WER。现在的路线也从“llm接受emg_emb”转为“使用llm做rerank”。

这批次实验的目的是：判断emg signals往text方向靠拢后，llama是否可以还原出语句

E0. dGaddy CTC baseline

pretrained recognition_model -> CTC decode

目的：确认当前数据/切分下的可靠 WER 下界

E1. closedVocab_llama.py 原始 adaptor baseline

raw/feature EMG -> adaptor -> frozen LLaMA

目的：复现实验管线

E2. dGaddy encoder frozen + semantic adaptor + frozen LLaMA

loss = CE

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文件：recognition_openVocab_llama3.py

目的：看 pretrained CTC hidden 是否比 raw adaptor 更适合 LLaMA

作为baseline

E3. dGaddy encoder frozen + semantic adaptor + frozen LLaMA

loss = CE + λ * cosine/InfoNCE alignment

目的：看“先让 EMG 靠近 text embedding”是否有帮助

E2E3 results:

LLaMA 输出更像语言模型续写，常常更流畅但离 EMG 更远。所以我们排除了：EMG continuous embedding -> LLaMA 自然利用语言先验

E4: E0 greedy -> LLaMA correction

看 LLM 是否能修 phonetic trace

E5: CTC forced alignment -> local segment-text InfoNCE probe

看局部 EMG segment 是否能对齐到 word/subword

E5 加强版：

给定 local EMG segment，能否检索到正确 word/subword/phoneme？

目的：证明 local EMG segment 是否含有可对齐到 word/subword/phoneme 的信息。

性质：probe，不直接作为最终 decoder。

关键：分桶报告 content word、rare word、长词、phoneme/subword top-k。

RESULTS:

phoneme top1/top5/top10: 0.087 / 0.305 / 0.491

word test-gallery top1/top5/top10: 0.810 / 0.927 / 0.947

word train-gallery top1/top5/top10: 0.739 / 0.825 / 0.852

ANALYSIS:

1. 当前 phoneme route 不稳，不能作为主线证据。

2. word-level local EMG evidence 非常强，即使用 train vocabulary，gold word 仍有 85.2% 概率出现在 top10。

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EMG hidden 不是完全没有语义/词级信息；

在局部 segment 上，它能提供很强的 word identity evidence。

E6: CTC n-best / local candidates + LLM reranking

看 LLM 作为受约束语言先验能否超过 E0 greedy

E6 加强版:

dGaddy-style CTC beam + LLaMA rerank

目的: 作为 dGaddy-style CTC+LM 的强 baseline。

性质: 不能当创新主线。

关键: 必须用 beam decode, 比较 KenLM / LLaMA rerank / oracle n-best。

RESULTS:

CTC beam top1 WER: 0.4283

CTC n-best oracle WER: 0.3879

LLaMA rerank WER: 0.5460

ANALYSIS:

这说明句级 LLaMA reranking 不但没帮忙, 还伤害结果。

它再次证明: LLM 如果只看整句候选, 缺少局部 EMG evidence 约束, 就会按语言概率选错。

E7:

目的: 提出真正不同于 dGaddy 的 constrained local evidence decoding。

性质: 主路线候选。

关键: 不是 rerank CTC 完整句子, 而是构造 segment-level phoneme/subword/word lattice, 再让 LLM 做受约束选择。

E7-word oracle-boundary 的作用是把 E5+ 的局部候选串成句子 lattice。

RESULTS:

CTC beam top1 WER: 0.4283

E7 lattice oracle WER: 0.1477

E7 decoded WER: 0.2549

并且逐句统计:

E7 decoded vs CTC beam: 80 better / 16 equal / 2 worse

在 oracle word boundary 下,

train-vocab local word candidates + constrained LLM 显著优于 CTC beam。

TODO:

local evidence constrained LLM 成立，但目前依赖 oracle boundary;

需要将 oracle boundary 替换成 predicted/local boundary。

tangmi0522周报

科研周报 | 姓名：汤宓 | 日期：2026-05-22 | 研究方向：语音在大脑中的表征

1.本周学习/研究内容

- 会议/讨论：
 - 参加了侵入式脑机接口课题组关于今后数据收集和课题组织的讨论。
 - 收获/进展：明确了华山的多模态数据分析思路。
- 文献阅读：
 - 阅读了文章：
 - 早期的bistable计算模型
 - Wilson HR (2003) Computational evidence for a rivalry hierarchy in vision. *J Comput Neurosci*
 - Laing CR, Chow CC (2002) A spiking neuron model for binocular rivalry. *Proc Natl Acad Sci USA*
 - Gigante, G., Mattia, M., Braun, J., & Del Giudice, P. (2009). Bistable perception modeled as competing stochastic integrations at two levels. *PLoS computational biology*, 5(7), e1000430
 - 相对近期的神经科学视角：
 - Wang, M., Arteaga, D., & He, B. J. (2013). Brain mechanisms for simple perception and bistable perception. *Proceedings of the National Academy of Sciences*, 110(35). <https://doi.org/10.1073/pnas.1221945110>
 - 主题/重点：bistable perception spiking neuronal model key word: reciprocal/ mutual inhibition; rivaling representation; perceptual reversal thresholds; metastable attractor.
- 代码/开发：
 - 编写/修改了代码：编写了华山EEG+眼动+行为学分析的代码，用于处理和分析多模态数据；编写了bistable词边界实验的single neuron sorting代码，完成了所有原始数据的初步处理与整理。

2.本周学习/研究总结

- 实验发现/结果：
 - 完成了华山多通道数据的分析报告，并已提交给陈昭荣(); 完成了bistable实验全部原始数据的single neuron sorting



EEG-Behavior-Eyetracking 分析报
告.pdf
1.43MB

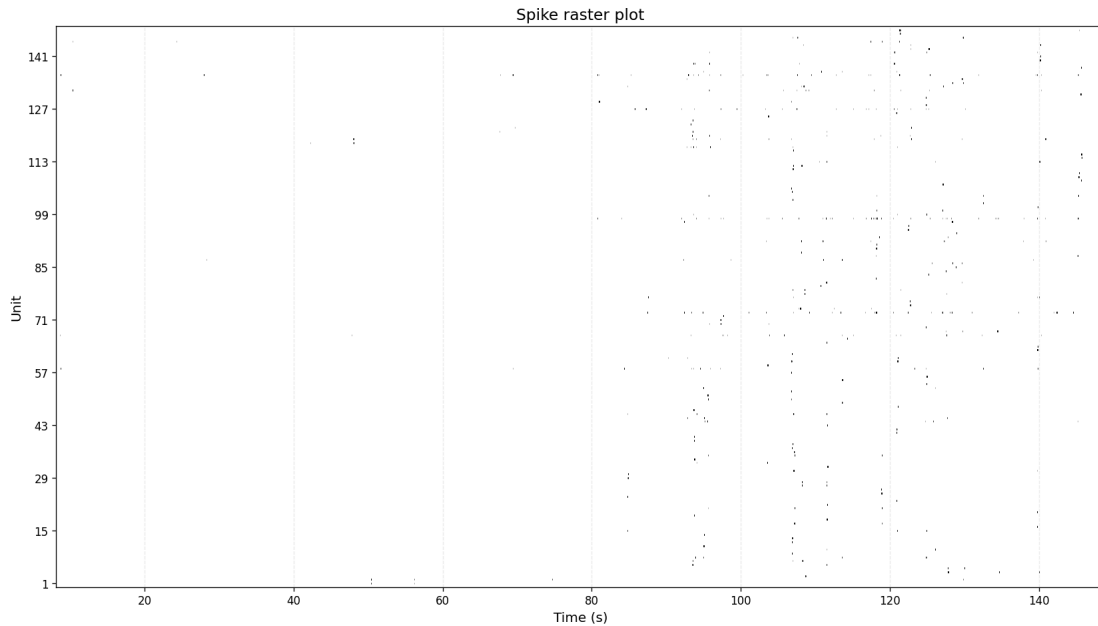


- 发现了问题：陈昭荣希望能够再更改图片命名范式的细节，但用于erp分析的trigger目前只有图片和发音前的2个时间点，可能需要在trigger上添加和优化更符合研究问题的时间点。
- 经验教训/失败分析：
 - 前期统一数据格式能够极大提升后续分析效率。

3.实验结果对比

- 与上周相比，本周结果有何变化/进展：从上周的代码编写与数据准备阶段，推进到了本周完成全部原始数据处理与导出sorting结果阶段。

index	sample	unit_id	time_sec					
0	258757	67	8.625233					
1	261952	136	8.731733					
2	261967	58	8.732233					
3	261982	136	8.732733					
4	261997	58	8.733233					
5	262037	58	8.734567					
6	262052	58	8.735067					
7	262077	58	8.7359					
8	262092	58	8.7364					
9	262147	136	8.738233					
10	262162	136	8.738733					
11	262177	136	8.739233					
12	262192	136	8.739733					
13	262217	136	8.740567					
14	262232	136	8.741067					
15	262247	136	8.741567					
16	262262	136	8.742067					
17	309748	145	10.32493					
18	309769	132	10.32563					
19	309849	132	10.3283					
20	309864	132	10.3288					
21	309969	132	10.3323					
22	309999	132	10.3333					
23	310039	132	10.33463					
24	310054	132	10.33513					
25	310254	132	10.3418					
26	310269	132	10.3423					
27	310284	132	10.3428					
28	310309	132	10.34363					
29	310324	132	10.34413					
30	310344	132	10.3448					
31	310359	132	10.3453					
32	310384	132	10.34613					
33	310409	132	10.34697					
34	310444	132	10.34813					
35	310479	132	10.3493					
36	310514	132	10.35047					
37	310544	132	10.35147					
38	310609	132	10.35363					
39	310624	132	10.35413					
40	310649	132	10.35497					
41	310664	132	10.35547					
42	310704	132	10.3568					
43	310749	132	10.3583					
44	310764	132	10.3588					
45	310794	132	10.3598					
46	310809	132	10.3603					
47	310824	132	10.3608					
48	310859	132	10.36197	50				
49	310894	132	10.36313					
50	310939	132	10.36463					



上图为来自ATL的spike sorting raster plot。148 个神经元（其中或许包含多个多神经元，之后需要通过调整threshold进一步分离单神经元），每个点代表一次 spike。图中可以看到80 s 之后多个单元同步活跃。

4.下周学习/研究计划

- 实验计划与改进：
 - 基于本周已完成排序的bistable实验数据，进行正式分析，形成初步实验结果。
- 文献阅读计划：
 - 需要阅读关于 single neuron analysis and interpretation protocol的文献。

5.其他事项/需求 (可选)

- 正在撰写深圳市基础研究专项面上项目申请书。

张志齐 未提交

孙嘉璐★未提交

张台忍 未提交

Jiayuhan0523周报

科研周报 | 姓名：贾煜晗 | 日期：2026-05-23 | 研究方向：EEG持续学习

1.本周学习/研究内容

- 实验进展：
 - 下载并预处理了M3CVDataset数据集
- 其他：
 - 完成毕业设计答辩
 - 进行TIP期刊rebuttal

3.下周学习/研究计划

- 测试数据集M3CVDataset上模型的效果
- 针对acil方法设计创新点